SPARMANIK 4.0 · STAGE-BY-STAGE PLAYBOOK

NEW SOP *Melon* Book

Grow it one stage at a time.

Read only the page you're on.

Target 1.8–2.2 kg · brix 15+



🌱 Prep → Nursery → Grow → Harvest

💧 Set-&-forget feed & water

Sparmanik Farm · Kab. Simalungun • SPARMANIK 4.0 · 6 Oct 2026

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What’s new in Sparmanik 4.0

This is book v3.8 with the lessons of the 2026 crop built in. Where the two disagree, 4.0 follows 2026.

#	Topic	v3.8 said	Rule in 4.0	Why (2026)
D1	Silicon drench	Weekly HST 8–57	**HST 8, 15, 22, 29, 36 only**	Drench at HST 50 → fruit split next day
D2	Finish	Plain water from HST 72	**Ripen feed to the last fruit**	Brix 14.6–19 reached on Ripen feed
D3	Boroni in Sizing	800 g, optional	**None — Flower stock only**	Boron damage this crop
D4	Leaf sprays	Ca foliar to HST 55	**Ca HST 35–44 · none HST 45–59 · Ripen: MKP / Calcinit alternate days**	Netting forms HST 45–60
D5	Stock batch	50 L (10 days)	**25 L (5 days) normal · 1 & 10 days too**	Less waste when plans change

Three golden rules

1. **Boric acid is banned.** Boron only from Karate Plus Boroni, in Boyd’s pre-weighed kit. Boric acid deformed 19% of the 2026 fruit.
2. **After fruit set, no plain water.** No glass watering, no plain-water tank, no silicon. Each split fruit.
3. **Humidity ≥ temperature × 2.** Dry air shuts the leaves, stops sugar and splits fruit (page 56).

Iron: Fe-EDDHA only, never Fe-EDTA

The 30 Jul 2026 Grow stock used Fe-EDTA 13%. It stops feeding iron above pH 6.5; our run-off was 7.0–7.8 → pale new leaves. Every Stock B uses MIKRO Fe 6% EDDHA, 50 g per 50 L.

Where the book changed

Pages	Change
10 · 27 · 42	No Boroni in Sizing stock
12 · 13 · 19 · 26 · 28	Silicon stops at HST 36
13 · 29 · 31 · 34–36	Ripen feed to the last fruit, no plain-water finish
26 · 34–36	Calcium leaf spray HST 35–44 only
9 · 11 · 12 · 13	v4.6 timings: Trichoderma HST 0, 17, 38, 59 · Tri-Pholate HST 20, 28, 35
50 – 62 (new)	Product photo cards · cost per crop · 1 / 5 / 10-day batches · humidity · harvest & grading · growth checks · run-off & call Boyd · emergency flush · 2026 lessons · checklist

HOW THE CROP RUNS

The stages

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2 Prep — buffer coir, bags, house 6

T Transplant + Trichoderma drench 8

M Make Stock A & B (night before) 14

3 Establish & grow (HST 0–25) 16

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4 Flower & set (HST 25–42) 22

5 Fruit sizing (HST 42–60) 26

6 Ripen & finish (HST 60–80) 29

7 Harvest & turnaround 32

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How to use this book

Find today's stage. Read *only* that page — it has what to do, feed & water, spray, what to watch for, and when to move on. The **Reference** at the back (exact grams, stock-making, ppm, checks, pictures) is for whoever makes up the drums.

START HERE · WHAT MOST DAYS NEED

Make the feed

A
5 L Stock A
(orange)

+

B
5 L Stock B
(blue)

=

1000 L
fill with water

Then check: **EC** = this week's number · **pH 5.8–6.3** · **Done** ✓

Grow → EC 2.1 **Flower → 2.4** **Sizing → 2.55** **Ripen → 2.7**

EACH MIX AUTO-DELIVERS THESE PPM — YOU DON'T SET THEM

	Grow	Flower	Sizing	Ripen
N	180	175	158	138
K	195	240	285	325
Ca	165	195	195	170

Full N-P-K-Ca-Mg table in Reference. K climbs to build sugar; Ca stays high against BER.

Fruiting & ripening only: also tip in **this week's SOP bag** (green) before topping up.

Always **5 L Stock A + 5 L Stock B** per 1000 L. Drums are pre-made for the stage, so the feed is always right — add the liquids, tip the SOP bag if it's a fruiting stage, top up, set EC + pH. No calculating.

STAGE 1

Nursery — seed to seedling

7–14 days · before HST 0

GERMINATE AT	SEEDLING EC	READY AT
28–30 °C	**0.5 → 1.5**	**2–3 true leaves**

1 · Soak the seed

1. **Warm clean water ~28 °C, 7–9 hours.** This "hydropriming" makes sprouting faster and more even. **Don't go past ~12 h** — seeds need oxygen or they drown.
2. **Optional but recommended (you have H_2O_2):** first dip seed in **~1% hydrogen peroxide for 10–15 min**, then rinse, then do the warm-water soak. It sanitises the seed coat (less damping-off) and lifts vigour — both research-backed.
3. Only soak fresh seed you will sow the same day.

✓ **Also supported**

Warm-water soak = quicker, more uniform germination. A dilute KNO_3 soak helps in *cold* weather — not needed in your warm nursery, so plain warm water (± the H_2O_2 dip) is all you need.

2 · Pre-sprout in paper towel

Damp (not soggy) towel, sealed in a box, **warm & dark 28–30 °C**. Radicle "legs" show in 2–4 days. Move each seed the moment the root is 5–10 mm — don't let roots grow into the towel.

3 · Into rockwool

Pre-soak cubes at **pH 5.8, EC 0.5**. Set the sprout **radicle-down**, ~1 cm deep. **Top-feed** to keep moist; dome on; 28–30 °C.

4 · Head breaks → light + bottom-feed

When cotyledons pop up: give **strong light immediately** (or seedlings go leggy) and dome off. **Switch to bottom-feed** (fill the tray, cube wicks up) — keeps the surface dry against damping-off.

5 · Grow on & harden

Bottom-feed, raise EC to ~1.0–1.5, pH 5.8–6.2, 25–28 °C day / 20–22 °C night, plenty of light. Harden off 2–3 days before transplant (a little less water, more light/air).

Watch for this stage

- ! **Damping-off** (seedling rots at the base) — from wet, cold, still air → bottom-feed, keep warm, airflow, don't overwater.
- ! **Leggy seedlings** — too little light after the head breaks → light them hard.
- ! **Fungus gnats** — from soggy media → don't overwater.

- **Weak / uneven sprouts** — discard them; plant only strong, uniform seedlings.

Move on when: seedlings have 2–3 true leaves, sturdy and uniform → ready to transplant — buffer the coir (Prep) first.

STAGE 2

Prep — get everything ready

Before planting

JOB	SUBSTRATE	STOCK
Set up	Coir buffered	A & B made

1 · Cocopeat & bags

1. **Reused coir:** after the water + H_2O_2 wash, rinse well and **rest ~1 day** so peroxide can't burn roots. Check EC — aim under 1.0.
2. **Re-buffer every cycle:** soak/drench with calcium nitrate (~1 kg Calcinit / 1000 L), then drain. This stops the coir stealing calcium and flushes sodium.
3. Blend in ~20–30% fresh coir or perlite to keep it airy; discard coir from any diseased crop.
4. Fill polybags with drainage holes; stand them on lifters ~1 inch off the floor; one 2 L/hr dripper per bag.

> ⚠ **Never skip the re-buffer**
>
> Un-buffered coir + your humidity = blossom-end rot from day one.

How to re-buffer — step by step

You need: 3 × 1000 L tanks of water · 3 kg Calcinit total (1 kg per tank) · a cup for the edge pour.

Day 1

1. **Put the polybags back in the greenhouse** — each bag in its correct row and number (A01, A02 … J43), on lifters ~1 inch off the floor.
2. **Put one dripper in each bag** — push it into the top of the cocopeat; check it is not blocked.
3. **Mix the tank:** fill to 1000 L, add **1 kg Calcinit**, stir until fully dissolved.
4. **Run 1:** run the drippers until the tank is empty. Check water drips out the bottom of the bags — and check the bags at the **far end of every row**. If they aren't wet, tell the supervisor.
5. **Wait 2–3 hours** — this is when the calcium sticks to the cocopeat. Don't skip it.
6. **Run 2:** mix a new tank (1000 L + 1 kg Calcinit) and run until empty.

Day 2

1. **Run 3:** mix a new tank (1000 L + 1 kg Calcinit) and run until empty.

2. **Edge pour — the last job:** keep ~**300 ml of the same Calcinit solution** for each bag. With a cup, pour it slowly **around the outside edge of the cocopeat**, near the rim — **not** in the middle where the dripper already went.

> ↑ **Why the edge pour matters**
>
> The dripper only wets a cone under itself — the **edges and corners stay dry**. Roots grow there too. If the edges get no calcium, you get blossom-end rot on the fruit.

> ⚠ **Never rinse with plain water afterwards**
>
> Plain water washes the calcium back out and ruins the whole job. **Then wait 2 days before transplanting** the seedlings.

2 · Ready the house

Weed-mat dry floor · fans running 24/7 · roof vents open · space bags for airflow · sanitise tools, drippers and structure before a new crop.

3 · Make the feed stock

Make up **Stock A** (orange) and **Stock B** (blue) for the stage — grams in **Reference**. Very strong — just **5 L of each per 1000 L**. For fruiting/ripening stages also pre-weigh the **SOP bags** (tipped straight into the tank).

Watch for this stage

- ! **Carry-over pests & disease** from the last crop → clean and sanitise everything; discard any bad coir.

Move on when: coir buffered (EC < 1.0), bags & drippers set, fans/vents running, house clean → ready to transplant, then Trichoderma drench.

TRANSPLANT DAY · ROOT PROTECTION

Trichoderma — Cup Drench

By cup, by hand, one bag at a time.

Transplant day

Gently transplant each seedling into its buffered polybag and firm the coir. Then do the Trichoderma drench below. **Plain water only today — no nutrient.** Nutrient feed starts tomorrow morning (HST 0).

NEVER put Trichoderma in the drippers or the 1000 L feed tank

Living spores in a powder — they block the 2 L/hr drippers and the filter, and the plants stop getting water. Do it by cup, by hand, one bag at a time.

1 · Mix it

1. Clean 10 L bucket. Fill with **PLAIN water only** — no nutrient.
2. Add the Trichoderma. Stir well for 1 minute.
3. Stand 15–30 min in the shade, then stir again.
4. Use it the **SAME DAY**. Never keep overnight. Never leave the bucket in the sun.

Dose: 1 gram per 1 litre → 10 grams in the 10 L bucket

Weigh it. Do not guess.

2 · Pour it

1. Water the bags normally **FIRST**, so the coir is already damp.
2. Pour **ONE CUP around the base of the stem**, in a circle. Do not splash the leaves.
3. Stir the bucket again every 10 bags, or the powder sinks to the bottom.
4. After the drench, do not run a big irrigation for the rest of the day, or it washes out.

Amounts		
ONE CUP **250 mL** per bag	ONE BUCKET **10 L** = 40 bags	WHERE **BASE** stem base

Never do this

- Never in the drippers or the feed tank.
- Never mix with H_2O_2 (peroxide), chlorine, fungicide, or any chemical — they kill it.
- Never on the same day as the Kalsika silicon drench — leave 2 days between.

When to do it

Time of day	Early morning or late afternoon only. Never in strong sun.
How often	HST 0 (transplant day), 17, 38 and 59 (v4.6). Write the date on the board.
After H_2O_2	Coir must rest 1 day first.

If unclear, STOP and ask the supervisor.

PRE-WEIGHED BAGS · ONE PER STAGE

One bag per stage

Everything is already in the bag. Nothing to add at mixing time.

The whole job

Tip **BAG A** into **Drum A**. Tip **BAG B** into **Drum B**. Top both to **50 L** and stir until clear. That is all.

	BAG GROW-A → DRUM A		BAG GROW-B → DRUM B		SOP PER TANK
GROW Vegetative HST 0–25 · 2 sets	Calcinit Yara	8.70 kg	Krista MKP Yara	2.20 kg	none
	Krista K Plus Yara	3.45 kg	MAG-S Meroke	5.70 kg	
	bag weight	12.15 kg	Vitaflex Meroke	0.20 kg	
			Micro Fe EDDHA 6% Meroke	0.05 kg	
			bag weight	8.15 kg	
FLOWER Generative HST 25–42 · 3 sets	Calcinit Yara	9.45 kg	Krista MKP Yara	2.30 kg	310 g
	Karate Plus Boroni Meroke	0.80 kg	MAG-S Meroke	6.05 kg	
	Krista K Plus Yara	1.15 kg	Vitaflex Meroke	0.20 kg	
	bag weight	11.40 kg	Micro Fe EDDHA 6% Meroke	0.05 kg	
			bag weight	8.60 kg	
SIZING Generative HST 42–60 · 3 sets	Calcinit Yara	9.45 kg	Krista MKP Yara	2.10 kg	535 g
	Boroni — not used (4.0) Meroke	—	MAG-S Meroke	6.25 kg	
	bag weight	9.45 kg	Vitaflex Meroke	0.20 kg	
			Micro Fe EDDHA 6% Meroke	0.05 kg	
			bag weight	8.60 kg	
RIPEN Generative HST 60–72 · 1 set	Calcinit Yara	8.95 kg	Krista MKP Yara	1.85 kg	645 g
	bag weight	8.95 kg	MAG-S Meroke	6.25 kg	
			Vitaflex Meroke	0.20 kg	
			Micro Fe EDDHA 6% Meroke	0.05 kg	
			bag weight	8.35 kg	

Three rules

1. A bag marked **A** only ever goes in **Drum A**; a bag marked **B** only ever in **Drum B**. Never swap them. **2.** One set = one Drum A + one Drum B = **ten 1000 L tanks**, at 5 L + 5 L per tank. **3. SOP is never in a bag** — weigh it fresh into each 1000 L tank, using the green column above.

One crop = 9 sets, 18 bags. Weigh out: Calcinit 83.05 kg · MAG-S 54.55 · SOP 28.20 (loose) · Krista MKP 19.45 · Krista K Plus 10.35 · Karate Plus Boroni 2.40 · Vitaflex 1.80 · Micro Fe EDDHA 0.45. Double-bag and seal the Calcinit — it pulls water from the air and sets hard.

NUTRITION · DAY BY DAY

Day by day — HST 0 to 24 • Grow

Find today. Read across: which bags, what strength, how much water, when, and the job due.

GROW FLOWER SIZING RIPEN

HST	BAG SET	EC MS	TDS PPM	SOP /1000 L	WATER PER POLYBAG ML	PULSES	WATERING TIMES	JOB FOR THE DAY
0	GROW	2100	1050	—	500	3 × 5m	8am · 12pm · 4pm	TRANSPLANT + Trichoderma cup drench
1	GROW	2100	1050	—	500	3 × 5m	8am · 12pm · 4pm	
2	GROW	2100	1050	—	500	3 × 5m	8am · 12pm · 4pm	
3	GROW	2100	1050	—	500	3 × 5m	8am · 12pm · 4pm	
4	GROW	2100	1050	—	500	3 × 5m	8am · 12pm · 4pm	
5	GROW	2100	1050	—	500	3 × 5m	8am · 12pm · 4pm	
6	GROW	2100	1050	—	500	3 × 5m	8am · 12pm · 4pm	
7	GROW	2100	1050	—	500	3 × 5m	8am · 12pm · 4pm	Start removing side shoots, every 2–3 days
8	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	Silicon drench — 2 pulses only today · TIMER CHANGE — 4 pulses from tomorrow
9	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	
10	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	
11	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	
12	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	
13	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	
14	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	
15	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	Silicon drench — 2 pulses only today
16	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	
17	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	Trichoderma cup drench
18	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	
19	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	
20	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	Node 8 — stop shoots · Tri-Pholate after 4 pm
21	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	
22	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	Silicon drench — 2 pulses only today
23	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	
24	GROW	2100	1050	—	800	4 × 6m	8 · 11am · 2 · 5pm	Mix the new FLOWER stock TONIGHT

pH 5.8 – 6.3 every day, every stage. Dripper 2 L/hr, one per polybag, two plants share it — every water figure is **per polybag**. On a silicon day run only the two afternoon pulses. SOP goes straight into the 1000 L tank, never into a bag. Check run-off 15–20%, two or three times a week, never on a silicon day.

NUTRITION · DAY BY DAY

Day by day — HST 25 to 49 • Flower into Sizing

The feed steps up twice here. Mix the new stock the night before each change.

GROW FLOWER SIZING RIPEN

HST	BAG SET	EC MS	TDS PPM	SOP /1000 L	WATER PER POLYBAG ML	PULSES	WATERING TIMES	JOB FOR THE DAY
25	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	CHANGE TO FLOWER MIX · start SOP · hand-pollinate 8–10 am
26	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	
27	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	
28	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	Tri-Pholate spray after 4 pm
29	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	Silicon drench — 2 pulses only today
30	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	
31	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	
32	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	
33	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	
34	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	
35	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	Tri-Pholate (last) · start Ca foliar
36	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	Silicon drench — 2 pulses only today
37	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	
38	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	Keep ONE fruit/plant · Trichoderma drench
39	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	
40	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	
41	FLOWER	2400	1200	310 g	1167	5 × 7m	8·10·12·2·4	Mix the new SIZING stock TONIGHT
42	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	
43	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	No silicon from HST 38 (4.0)
44	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	
45	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	Netting starts — no leaf spray from here
46	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	
47	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	
48	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	
49	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	

pH 5.8 – 6.3 every day, every stage. Dripper 2 L/hr, one per polybag, two plants share it — every water figure is **per polybag**. On a silicon day run only the two afternoon pulses. SOP goes straight into the 1000 L tank, never into a bag. Check run-off 15–20%, two or three times a week, never on a silicon day.

NUTRITION · DAY BY DAY

Day by day — HST 50 to 75 • Sizing to harvest

4.0: no silicon after Day 36 · Ripen feed to the last fruit, no plain-water finish.

GROW FLOWER SIZING RIPEN

HST	BAG SET	EC MS	TDS PPM	SOP /1000 L	WATER PER POLYBAG ML	PULSES	WATERING TIMES	JOB FOR THE DAY
50	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	No silicon (4.0) — splits fruit
51	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	
52	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	
53	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	
54	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	
55	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	
56	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	
57	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	No silicon (4.0) — splits fruit
58	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	
59	SIZING	2550	1275	535 g	1600	6 × 8m	7·9·11·1·3·5	Mix RIPEN stock TONIGHT · Trichoderma
60	RIPEN	2700	1350	645 g	800	4 × 6m	8·11am·2·5pm	CHANGE TO RIPEN MIX
61	RIPEN	2700	1350	645 g	800	4 × 6m	8·11am·2·5pm	
62	RIPEN	2700	1350	645 g	800	4 × 6m	8·11am·2·5pm	
63	RIPEN	2700	1350	645 g	800	4 × 6m	8·11am·2·5pm	
64	RIPEN	2700	1350	645 g	800	4 × 6m	8·11am·2·5pm	
65	RIPEN	2700	1350	645 g	800	4 × 6m	8·11am·2·5pm	
66	RIPEN	2700	1350	645 g	800	4 × 6m	8·11am·2·5pm	
67	RIPEN	2700	1350	645 g	800	4 × 6m	8·11am·2·5pm	
68	RIPEN	2700	1350	645 g	800	4 × 6m	8·11am·2·5pm	
69	RIPEN	2700	1350	645 g	800	4 × 6m	8·11am·2·5pm	
70	RIPEN	2700	1350	645 g	800	4 × 6m	8·11am·2·5pm	
71	RIPEN	2700	1350	645 g	800	4 × 6m	8·11am·2·5pm	
72	RIPEN	2700	1350	645 g	800	4 × 6m	8 · 11am · 2 · 5pm	Ripen feed to the last fruit · rolling pick
73	RIPEN	2700	1350	645 g	800	4 × 6m	8 · 11am · 2 · 5pm	
74	RIPEN	2700	1350	645 g	800	4 × 6m	8 · 11am · 2 · 5pm	
75	RIPEN	2700	1350	645 g	800	4 × 6m	8 · 11am · 2 · 5pm	

pH 5.8 – 6.3 every day, every stage. Dripper 2 L/hr, one per polybag, two plants share it — every water figure is **per polybag**. On a silicon day run only the two afternoon pulses. SOP goes straight into the 1000 L tank, never into a bag. Check run-off 15–20%, two or three times a week, never on a silicon day.

THE NIGHT BEFORE HST 0 · S1 STOCK

Making Stock A & Stock B

50 litre drum · weigh every product.

DRUM A — orange label · 50 L · 2 products only

1	**CALCINIT** Yara Calcinit · calcium nitrate	**8.7 kg** 8 kg 700 g
2	**K PLUS** Yara Krista K Plus · potassium nitrate	**3.45 kg** 3 kg 450 g

DRUM B — blue label · 50 L · 4 products

1	**MKP** Yara Krista MKP · mono potassium phosphate	**2.2 kg** 2 kg 200 g
2	**MAG-S** Meroke MAG-S · magnesium sulphate	**5.7 kg** 5 kg 700 g
3	**VITAFLEX** Meroke Vitaflex · pink bag · use kitchen scale	**200 g** 0.2 kg
4	**Fe-EDDHA** iron chelate · use kitchen scale	**50 g** 0.05 kg

⚠ **Weigh every product. Do not guess.**

S1 FEED · DRUM

How to fill a drum

1. Fill the drum **HALF FULL** with warm water.
2. Weigh product 1. Pour it in. Stir until the water is **CLEAR**.
3. Only when clear, weigh the next product. Pour in. Stir until clear.
4. Repeat until every product is in the drum.
5. Add water up to the **50 LITRE** mark.
6. Write **A or B** and **TODAY'S DATE** on the drum.
7. Close the lid. Keep out of the sun. Use within 2 weeks.

⚠ Never do this

- Never put a Drum A product into Drum B, or the reverse.
- Never mix Drum A and Drum B together — they meet only in the 1000 L tank.
- Never share a scoop or stirrer between A and B.
- Never add the next product while the water is still cloudy.
- Never put SOP or Kalsika into these drums.

↑ Using the drums

5 L from DRUM A + 5 L from DRUM B → 1000 L water. Water first → add A → stir → add B → stir → top up to 1000 L. Then set **EC 2.1**, check **pH 5.8–6.3**. Use a marked 5 L jug. One drum makes 10 tanks of 1000 L.

STAGE 3 · VEGETATIVE

Establish & grow

HST 0 – 25

FEED EC	WATER	DRENCH
2.1	**every 3 h**	**Silicon weekly**

Do

1. Transplant the seedling into buffered coir; settle the roots gently.
2. Train the vine up its string; remove side-shoots (laterals) **below node 8**; keep nodes 9–12 for fruiting later.
3. Start the **silicon root drench weekly** from ~HST 8 (Kalsika 0.3–0.5 g/L in plain water, its own pulse — never in the feed).

Feed & water

Feed: Stock A + Stock B → 1000 L · **EC 2.1** · pH 5.8–6.3 (no SOP bag yet). How-to: see the START HERE card.
Water: establish week — every 4 h, 5 min; then every 3 h, 6 min. Steer by run-off (~15–20%); change only at stage change.

Watch for this stage

- **Aphids (kutu daun)** — clusters under leaves, curling; they spread mosaic virus. Hit them early.
- **Thrips** — silvery flecks, curled young leaves; also a virus carrier.
- **Whitefly** — tiny white flies under leaves.
- **Spider mites (tungau)** — fine webbing, speckled leaves (worse when hot/dry).
- **Caterpillars (ulat)** — chewed leaves.
- **Early powdery / downy mildew** — white powder or yellow patches on leaves.

Do: check ~10 plants per row, 2×/week (look under leaves) · **act at first sighting — don't wait for spread** · yellow & blue sticky traps · prune/remove affected parts · rotate a registered insecticide/fungicide per label (follow the label's safety & re-entry time) · net sides help keep pests out. **First mildew spots or mite stippling → treat that day; keep fruit-fly traps up continuously.**

Move on when: first female flowers appear (~HST 25) → Stage 4. **Make the new S2 stock the night before.**

STAGE 3 · S1 FEED · EC 2.1 · PH 5.8–6.3

Watering — Day 0 to Day 25

Two timer settings only.

DAY 0–7 — Establish / root settling

EVERY	MINUTES	PER PLANT
4 h	5	0.5 L

8 am · 12 pm · 4 pm

▼ ON DAY 8, CHANGE THE TIMER ▼

DAY 8–25 — Grow

EVERY	MINUTES	PER PLANT
3 h	6	0.8 L

8 am · 11 am · 2 pm · 5 pm

Rules

- Set the timer **ONCE**. Leave it until Day 8.
- Do NOT change the timer because it rained — the coir drains itself.
- Never water at night. All pulses in daylight.
- From Day 8: silicon drench (Kalsika) **once a week**, plain water, on its own — never in the feed tank.

STAGE 3 · RUN-OFF

Check run-off 2–3× a week

Target 15–20% drains from the bottom.

If…	Do
Too little under 15%	Add 1 minute to every pulse.
Too much over 20%	Remove 1 pulse from the day.

⚠ **Important**

Only change it if it is wrong for **several days** — not for one day.

Run-off log

Write the date, day (HST), drain %, and your name — 2–3× each week.

Date	HST	EC	pH	Drain %	Name

Day 25: first female flowers appear → move to Stage 4, change the feed and the timer. **Mix the new S2 stock the night before.**

FROM DAY 8 · ONCE A WEEK · ROOT DRENCH

Silicon (Kalsika) — Weekly Root Drench

Kalsika = potassium silicate, 100 g sachets. To the roots, plain water only. Weekly from Day 8, **stop at Day 36 (4.0).**

What it does

Silicon builds into the cell walls: **stronger stems, thicker tougher leaves** — aphids, thrips, mites and mildew struggle to get in. Also helps with heat and calcium.

Do it step by step — MANUAL switch

Fresh plain water only — no Stock A, no Stock B, no SOP. Use the manual switch for all three runs; do not change the automatic timer.

1. **Fill the tank with fresh water to 400 L.** Run **3 minutes** — this pushes the old feed out of the lines.
   → **about 300 L should be left in the tank. If not, stop and tell the supervisor.**
2. **Put 1 sachet (100 g) of Kalsika straight into the 300 L in the tank.** Do not dissolve it in a bucket first. Do not add more water. Stir it **in the tank** until the water is **CLEAR** — still cloudy, keep stirring, do not switch on.
3. **Run 8 minutes** — this waters every polybag.
   → **about 30 L left in the tank.**
4. **Top up with fresh water to 100 L only** — not more. Run **3 minutes** until the tank is empty — this pushes the Kalsika out of the lines.
5. **Make the normal feed:** fill the tank to **1000 L**, add **5 L Stock A + 5 L Stock B**, stir, set the stage **EC**, check **pH 5.8–6.3**. Write the date on the board.

On a silicon day, set the timer to 2 pulses only

The three runs already give the bags their morning water. So on silicon day run **only the afternoon pulses — 2 pm and 5 pm**. Next morning, back to the normal 4 pulses.

If the drippers cannot be used: 3 g per 10 L bucket, **250 ml at the base of each polybag** (40 bags per bucket, half a day for the farm).

Never do this

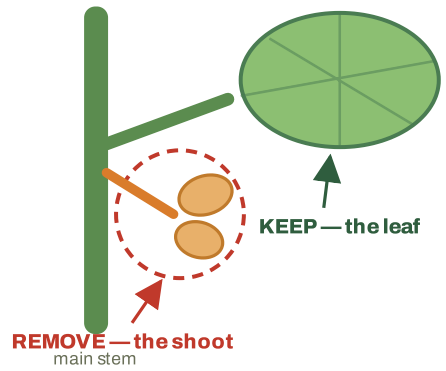
- **Drum A and Drum B — never, for any reason.** In the concentrate drums Kalsika sets into white sludge.
- **The 1000 L tank — one at a time.** Empty of Stock A, B and SOP before Kalsika goes in; empty of Kalsika before the feed goes back. Runs 1 and 3 keep them apart.
- Never the same day as Trichoderma — 2 days between.

STAGE 3 · VEGETATIVE · FROM HST 7

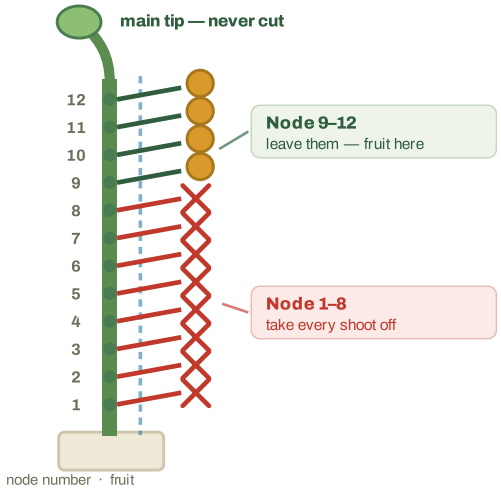
Remove the side shoots

Tunas air — the little shoots in the leaf armpits. Every 2–3 days until node 8.

1 · What to pinch out



2 · How far up the plant



How to do it

1. Find the small shoot growing in the **armpit** — where the leaf stalk meets the main stem.
2. **Pinch it out with your fingers** while it is small and soft. Do not pull hard, or the stem tears.
3. **Take the SHOOT only. Leave the LEAF.** The leaf is feeding the plant.
4. Do **not** cut the main growing tip. The leader must keep climbing the string.
5. Work in the **morning**, when leaves are dry, so the wound closes before night.
6. **No shoots on a plant yet? Leave it alone** — do not go looking or pulling. Check again in **2–3 days**. New shoots keep coming until node 8, around Day 20.

↑ Where to stop

Strip every shoot from node 1 to node 8. From **node 9 upward, leave the shoots alone** — the female flowers, and our fruit, come from those.

⚠ Never do this

Never remove a leaf — a young plant has only 4–6 and needs them all. **Never cut the top tip.** If you cannot tell which is the leader, stop and ask the supervisor. Do not do this on a **leaf-spray** day. The silicon drench goes to the roots, not the leaves — silicon and shoot removal on the same morning is fine.

STAGE 3 · VEGETATIVE · HST 8 – 25

Carry on the same until Day 25

Nothing changes now. Just repeat the weekly routine below.

↑ The feed does NOT change

S1 Grow — 5 L Stock A + 5 L Stock B per 1000 L. Set **EC 2.1**, check **pH 5.8–6.3**. **No SOP yet** — SOP only starts at Flower. Make a fresh tank whenever it runs out.

↑ The timer does NOT change

Every **3 hours, 6 minutes** — 8 am · 11 am · 2 pm · 5 pm. Leave it alone until Day 25. The only exception is silicon day: **2 pulses only**, 2 pm and 5 pm.

The weekly routine

Job	How often
Silicon drench (Kalsika)	**Once a week** — see the page before this one
Check run-off is 15–20%	**2–3 times a week** — not on a silicon day
Trichoderma cup drench	Every **2–3 weeks** — 2 days away from silicon
Train the vine, remove laterals below node 8	As the plants grow
Look under the leaves for pests	**2 times a week** — check ~10 plants per row
Write EC, pH, drain % and your name in the log	Every time you check

⚠ Do not change anything on your own

Do not change the feed, the EC, or the timer because of rain, one dry-looking day, or one odd reading. Only adjust if run-off is wrong for **several days in a row**. If something looks wrong, **tell the supervisor**.

Around Day 25 the first female flowers appear — small melon behind the petals. That is the signal to move to **Stage 4 · Flower**. **Mix the new S2 stock the night before.** Turn the page.

STAGE 4 · GENERATIVE

Flower, pollinate & set

HST 25 – 42

FEED EC	WATER	BORON
2.4	**every 2 h**	**via drip (Boroni)**

Do

1. **Hand-pollinate 8–10 a.m.** on nodes 9–12 — touch the male flower into the female. Do it daily through flowering (wet/humid air makes natural pollination poor).
2. Let a few set, then **keep ONE best fruit** per plant (~HST 38–40) and remove the rest.
3. **Top the plant** (stop the leader) at ~25 leaves to push energy into the fruit.

Feed & water

Feed: Stock A + Stock B + **SOP bag** → 1000 L · **EC 2.4** (boron rides in on the Boroni — no boron spray).
Water: every 2 h, 7 min. **Spray:** silicon weekly; start **calcium foliar on the young fruit ~HST 35** (Ca-nitrate 4–5 g/L, morning) — preventive because the house stays humid.

Watch for this stage

- **Thrips** — worst at flowering; deform flowers & young fruit, carry virus.
- **Fruit flies (lalat buah)** — hang methyl-eugenol traps **now**, before fruit forms.
- **Poor pollination** — humid, still air → hand-pollinate, keep fans on.
- **Gummy Stem Blight** — brown gummy ooze at the stem base in hot+humid air; can wilt/kill the plant. Watch the stem.
- **Powdery / downy mildew, aphids.**

Move on when: one fruit is set and starting to swell → Stage 5. **Make the new S3 stock the night before.**

THE NIGHT BEFORE DAY 25 · S2 STOCK

New Stock A & B — Flower

50 litre drum · weigh every product · feed EC 2.4.

↑ Make this the night before Day 25

Mix the new Stock A and Stock B **the evening before**, so the Flower feed is ready first thing in the morning. Do not leave it until the morning of the change.

DRUM A — orange label · 50 L · 3 products

1	**CALCINIT** Yara Calcinit · calcium nitrate	**9.45 kg** 9 kg 450 g
2	**BORONI** Karate Plus Boroni · boron	**800 g** 0.8 kg
3	**K PLUS** Yara Krista K Plus · potassium nitrate	**1.15 kg** 1 kg 150 g

DRUM B — blue label · 50 L · 4 products

1	**MKP** Yara Krista MKP · mono potassium phosphate	**2.3 kg** 2 kg 300 g
2	**MAG-S** Meroke MAG-S · magnesium sulphate	**6.05 kg** 6 kg 50 g
3	**VITAFLEX** Meroke Vitaflex · pink bag · use kitchen scale	**200 g** 0.2 kg
4	**Fe-EDDHA** iron chelate · use kitchen scale	**50 g** 0.05 kg

↑ SOP starts now — separate bag, NOT in the drum

Pre-weigh **310 g of SOP per 1000 L** into one bag per tank. Dissolve the bag in a bucket of warm water, then pour it into the 1000 L tank. Never put SOP in Drum A or Drum B.

⚠ Weigh every product. Do not guess.

Same method as the first drum: half-fill with warm water, add **one product at a time** and stir until the water is **CLEAR** before the next one, then top up to the **50 L** mark. Write **A** or **B** and today's date on the drum. Never mix A and B together.

STAGE 4 · GENERATIVE · S2 FEED · EC 2.4

Watering — Day 25 to Day 42

One timer setting for the whole stage.

DAY 25–42 — Flower & fruit set

EVERY	MINUTES	PER PLANT
2 h	**7**	**1.2 L**

8 am · 10 am · 12 pm · 2 pm · 4 pm

Rules

- Set the timer **ONCE** on Day 25. Leave it until Day 42.
- Do NOT change the timer because it rained — the coir drains itself.
- Never water at night. All pulses in daylight.
- Keep the midday pulses — that is when the plant drinks most.
- Silicon drench (Kalsika) **once a week**, plain water, on its own.

↑ **Steer by run-off**

Check 2–3× a week, aim **15–20%** drain. Under 15% → add **1 minute** to every pulse. Over 20% → remove **1 pulse** from the day. Only change it if it is wrong for **several days**.

Day 42: fruit set and starting to swell → Stage 5, change the feed and the timer. **Make the new S3 stock the night before.**

STAGE 4 · GENERATIVE · HAND POLLINATION

Hand Pollination

Use the plant's own flowers first. Early morning.

First choice: the plant's own flower

Take the male flower from the **SAME plant** as the female flower. This is the normal way. Do it this way whenever you can.

How to do it

1. Work **early morning** (about 6–9 am), when the flowers are freshly open.
2. Find an open **female** flower — it has a tiny melon behind the petals. A **male** flower has a thin stem and no melon.
3. Take a fresh male flower **from that same plant**. Peel the petals back to show the pollen.
4. Brush the pollen gently onto the centre of the female flower. One female = one gentle dab.
5. Do **not** use one flower to pollinate many plants. Use a fresh male flower each time.

If the plant has no male flower open that day

Use the **nearest plant of the SAME variety and the SAME generation** — nothing else.

- **F1 flower → F1 plant only. F3 flower → F3 plant only.**
- Never put F2 or F3 pollen on an F1 plant, or F1 pollen on an F2 or F3 plant.
- Never mix varieties — for example, do **not** use Yellow Kirin Australia F3 (row J) on the Kevin lines.

Write it on the tag

If you used a flower from **another plant** and the fruit sets, write it on the tag using the plant IDs — for example **H18A × H20B**. If it was the plant's own flower, write **self**. This is how we know which seed to keep.

Why it matters

Mixing generations or varieties mixes the genetics of the **seed**. Seed saved from a wrong cross gives uneven plants and fruit next season — different look, size and taste. Keeping it clean keeps the variety **true to type**.

STAGE 5 · GENERATIVE

Fruit sizing

HST 42 – 60

FEED EC	WATER	FRUIT
2.55	**every 2 h**	**1 per plant**

Do

1. **Support the fruit** in a sling/net tied to the wire so the stalk isn't strained.
2. Lightly remove a few old lower leaves for airflow; the fruit's netting develops now.

Feed & water

Feed: Stock A + Stock B + **SOP bag** → 1000 L · **EC 2.55**.
Water: every 2 h, 8 min — peak. **Spray:** no leaf spray, no silicon (netting · 4.0).

Watch for this stage

- **Blossom-end rot (BER)** — sunken brown patch on the fruit base = calcium (your #1 risk). See the picture guide in Reference. Fix: airflow, steady water, don't over-push K.
- **Fruit flies** — sting young fruit → maggots & rot inside. Keep traps up; consider **bagging each fruit**.
- **Gummy Stem Blight, powdery/downy mildew, anthracnose, fruit rot.**
- **Spider mites, aphids.**

Move on when: fruit near full size and netting complete (~HST 60) → Stage 6. **Make the new S4 stock the night before.**

THE NIGHT BEFORE DAY 42 · S3 STOCK

New Stock A & B — Sizing

50 litre drum · weigh every product · feed EC 2.55.

↑ Make this the night before Day 42

Mix the new Stock A and Stock B **the evening before**, so the Sizing feed is ready first thing in the morning. Do not leave it until the morning of the change.

DRUM A — orange label · 50 L · 2 products

1	**CALCINIT** Yara Calcinit · calcium nitrate	**9.45 kg** 9 kg 450 g
2	**BORONI** NOT used in 4.0 — boron in Flower only	— —

DRUM B — blue label · 50 L · 4 products

1	**MKP** Yara Krista MKP · mono potassium phosphate	**2.1 kg** 2 kg 100 g
2	**MAG-S** Meroke MAG-S · magnesium sulphate	**6.25 kg** 6 kg 250 g
3	**VITAFLEX** Meroke Vitaflex · pink bag · use kitchen scale	**200 g** 0.2 kg
4	**Fe-EDDHA** iron chelate · use kitchen scale	**50 g** 0.05 kg

Not used this stage: K Plus or Boroni (4.0). Drum A is Calcinit only.

↑ SOP bag — separate, NOT in the drum

Pre-weigh **535 g of SOP per 1000 L** into one bag per tank. Dissolve it in a bucket of warm water, then pour into the 1000 L tank.

⚠ Weigh every product. Do not guess.

Same method as the first drum: half-fill with warm water, add **one product at a time** and stir until the water is **CLEAR** before the next one, then top up to the **50 L** mark. Write **A** or **B** and today's date on the drum. Never mix A and B together.

STAGE 5 · GENERATIVE · S3 FEED · EC 2.55

Watering — Day 42 to Day 60

The wettest stage — six pulses a day.

DAY 42–60 — Fruit sizing

EVERY	MINUTES	PER PLANT
2 h	**8**	**1.6 L**

7 am · 9 am · 11 am · 1 pm · 3 pm · 5 pm

Rules

- Set the timer **ONCE** on Day 42. Leave it until Day 60.
- Do NOT change the timer because it rained — the coir drains itself.
- Never water at night. All pulses in daylight.
- Keep the midday pulses — that is when the plant drinks most.
- This is the heaviest watering of the crop — never let the bag dry out while the fruit is swelling.
- No silicon drench **in this stage** — after fruit set it splits fruit (4.0).

↑ **Steer by run-off**

Check 2–3× a week, aim **15–20%** drain. Under 15% → add **1 minute** to every pulse. Over 20% → remove **1 pulse** from the day. Only change it if it is wrong for **several days**.

Day 60: fruit near full size, netting complete → Stage 6, change the feed and the timer. **Make the new S4 stock the night before.**

STAGE 6 · GENERATIVE

Ripen & brix finish

HST 60 – 80

FEED	WATER	TEST
S4 →to the last fruit	**steady**	**brix ≥ 15**

Do

1. Feed the ripening mix (low N, high K) **to the last fruit** — no switch to plain water (4.0). Pick ripe fruit every morning; leave the rest on the feed to sweeten.
2. **Keep feed and timer steady to harvest.** 2026: Brix 14.6–19 on Ripen feed alone, while plain water after fruit set split fruit. Never add water by hand or glass.
3. Start **brix testing** (~HST 68) with the refractometer.
4. Keep water **steady**— sudden water = cracked fruit. Leaf spray 16:00: MKP / Calcinit, alternate days.

✓ **4.0 rule — no plain-water finish**

Ripen feed (EC 2700, SOP 645 g) runs until the last fruit is picked.

Brix test on fruit that split that day — never cut a good fruit.

Feed & water

Feed: Stock A + Stock B + **SOP bag** → 1000 L · **EC 2.7**, to the last fruit (4.0).
Water: every 3 h, 6 min →same to harvest.

Watch for this stage

- **Fruit flies — CRITICAL now.** They attack ripening fruit; keep traps, bag fruit, and destroy any stung fruit (don't leave it in the house).
- **Fruit cracking** — from water spikes; keep watering steady, roof keeps rain off.
- **Fruit rot / anthracnose, powdery mildew, Gummy Stem Blight** (can kill the plant before ripening → low brix).
- **Rodents / birds.**

Move on when: full slip + full net + aroma + **brix ≥ 15** → harvest.

THE NIGHT BEFORE DAY 60 · S4 STOCK

New Stock A & B — Ripen

50 litre drum · weigh every product · feed EC 2.7.

Make this the night before Day 60

Mix the new Stock A and Stock B **the evening before**, so the Ripen feed is ready first thing in the morning. Do not leave it until the morning of the change.

DRUM A — orange label · 50 L · 1 product only

1	**CALCINIT** Yara Calcinit · calcium nitrate	**8.95 kg** 8 kg 950 g

DRUM B — blue label · 50 L · 4 products

1	**MKP** Yara Krista MKP · mono potassium phosphate	**1.85 kg** 1 kg 850 g
2	**MAG-S** Meroke MAG-S · magnesium sulphate	**6.25 kg** 6 kg 250 g
3	**VITAFLEX** Meroke Vitaflex · pink bag · use kitchen scale	**200 g** 0.2 kg
4	**Fe-EDDHA** iron chelate · use kitchen scale	**50 g** 0.05 kg

Not used this stage: K Plus and Boroni. Drum A is Calcinit only.

SOP bag — separate, NOT in the drum

Pre-weigh **645 g of SOP per 1000 L** into one bag per tank. Dissolve it in a bucket of warm water, then pour into the 1000 L tank.

Weigh every product. Do not guess.

Same method as the first drum: half-fill with warm water, add **one product at a time** and stir until the water is **CLEAR** before the next one, then top up to the **50 L** mark. Write **A** or **B** and today's date on the drum. Never mix A and B together.

STAGE 6 · GENERATIVE · S4 FEED TO HARVEST

Watering — Day 60 to Harvest

4.0: one setting, Day 60 to the last fruit.

DAY 60–72 — Ripening · S4 feed · EC 2.7

EVERY	MINUTES	PER PLANT
3 h	6	0.8 L

8 am · 11 am · 2 pm · 5 pm

▼ 4.0: NO CHANGE ON DAY 72 — KEEP THE RIPEN FEED ▼

NOT USED — v3.8 finish, not used in 4.0

EVERY	MINUTES	PER PLANT
4 h	6	0.6 L

8 am · 12 pm · 4 pm

Rules

- From Day 72 the tank stays **RIPEN FEED** (EC 2700) — never plain water.
- Steady water + high-K feed lifts the sugar; dry-then-wet splits fruit.
- Never water at night. All pulses in daylight.
- No silicon in this stage (stops at Day 36).

↑ **Steer by run-off**

Aim **10–15%** drain in this stage — a little drier than before. If leaves wilt badly at midday, add 1 minute; otherwise leave it alone.

Harvest when: full slip + full net + aroma + **brix ≥ 15**.

STAGE 7

Harvest & turnaround

HST 75 – 85+

PASS	WEIGHT	THEN
brix ≥ 15	**1.8–2.2 kg**	**reset house**

Harvest

1. Signs: full netting · ground-spot yellowing · aroma at the stem · half-to-full slip.
2. Test one per batch — refractometer on mid-flesh juice (not the centre cavity). **Pass = brix ≥ 15, weight 1.8–2.2 kg.** Grade A–D and log it.
3. Pick in the cool morning; handle gently; leave a short T-stalk if storing.

Turnaround (back to Stage 2)

Remove the crop & debris → sanitise house, tools, drippers → wash + H_2O_2 the coir → rest → EC check → re-buffer with calcium nitrate. Then you're back at Stage 0.

Watch for

- **Carry-over pests/disease** — clean thoroughly; never reuse coir from a diseased crop.

YEAR PLAN · 3 CROPS

Crop schedule

Three crops a year. Find the week you're in and do that row. Dates follow the SOP — shift ±a few days to the real transplant date.
Daily always: irrigation + fertigation, EC + pH, pest scouting.

Crop 1 — seed 1 Apr → harvest Jul 2026

Week	HST	Stage	Do this week	EC
Nursery 1–14 Apr	−14→0	Nursery	**Seed soak 1 Apr.** Warm water 7–9 h → sprout → rockwool → bottom-feed. Harden. 2–3 leaves.	0.5→1.5
1 15–21 Apr	0–6	Establish	**Transplant ~15 Apr.** Settle roots. Water every 4 h→3 h. Train. Daily EC/pH.	2.1
2 22–28 Apr	7–13	Grow	Train vine. Remove laterals below node 8. Silicon drench weekly. Scout pests.	2.1
3 29 Apr–5 May	14–20	Grow	Keep training. Silicon weekly. Watch aphids/thrips/mildew.	2.1
4 6–12 May	21–27	Grow→Flower	First flowers (~HST 25). Feed EC 2.4 (Boroni in). Hand-pollinate 8–10 AM.	2.1→2.4
5 13–19 May	28–34	Flower/Set	Hand-pollinate daily 8–10 AM. Water every 2 h. Fruit-fly traps up.	2.4
6 20–26 May	35–41	Set/Select	**Keep ONE fruit/plant.** Top ~25 leaves. Start Ca foliar.	2.4
7 27 May–2 Jun	42–48	Sizing	Feed EC 2.55. Water every 2 h, 8 min. Support fruit. Ca foliar to HST 44.	2.55
8 3–9 Jun	49–55	Sizing	Netting forms. No leaf spray. Bag fruit/traps. Watch BER.	2.55
9 10–16 Jun	56–62	Sizing→Ripen	Fruit near full size. Prepare feed switch.	2.55→2.7
10 17–23 Jun	63–69	Ripen	Feed EC 2.7 (cut N, max K). Start brix testing. Steady water.	2.7
11 24–30 Jun	70–76	Ripen/Finish	Ripen feed continues. Rolling pick, ripe fruit only. Fruit fly CRITICAL.	2.7
12 1–8 Jul	77–84	Finish	Ripen feed. Brix ≥ 15. Watch cracking. Destroy stung fruit.	2.7
HARVEST 9–16 Jul	85–92	Harvest	Pick fully-ripe only (slip+net+aroma, brix ≥ 15). Sort, pack, sell now.	—

Turnaround (~mid–late Jul): harvest, remove plants, clean & disinfect, flush lines, re-buffer coir, make new stock. Gap to Crop 2 transplant (20 Jul) is tight.

Crop 2 — Jul–Oct 2026

Week	HST	Stage	Do this week	EC
Nursery 6–19 Jul	−14→0	Nursery	Soak 7–9 h → sprout → rockwool → bottom-feed. Harden. 2–3 true leaves.	0.5→1.5
1 20–26 Jul	0–6	Establish	**Transplant 20 Jul.** Settle roots. Water every 4 h→3 h. Train. Daily EC/pH.	2.1
2 27 Jul–2 Aug	7–13	Grow	Train vine. Remove laterals below node 8. Silicon drench weekly. Scout pests.	2.1
3 3–9 Aug	14–20	Grow	Keep training. Silicon weekly. Watch aphids/thrips/mildew.	2.1
4 10–16 Aug	21–27	Grow→Flower	First flowers (~HST 25). Feed EC 2.4 (Boroni in). Hand-pollinate 8–10 AM.	2.1→2.4
5 17–23 Aug	28–34	Flower/Set	Hand-pollinate daily 8–10 AM. Water every 2 h. Fruit-fly traps up.	2.4
6 24–30 Aug	35–41	Set/Select	**Keep ONE fruit/plant.** Top ~25 leaves. Start Ca foliar.	2.4
7 31 Aug–6 Sep	42–48	Sizing	Feed EC 2.55. Water every 2 h, 8 min. Support fruit. Ca foliar to HST 44.	2.55
8 7–13 Sep	49–55	Sizing	Netting forms. No leaf spray. Bag fruit/traps. Watch BER.	2.55
9 14–20 Sep	56–62	Sizing→Ripen	Fruit near full size. Prepare feed switch.	2.55→2.7
10 21–27 Sep	63–69	Ripen	Feed EC 2.7 (cut N, max K). Start brix testing. Steady water.	2.7
11 28 Sep–4 Oct	70–76	Ripen/Finish	Ripen feed continues. Rolling pick, ripe fruit only. Fruit fly CRITICAL.	2.7
12 5–12 Oct	77–84	Finish	Ripen feed. Brix ≥ 15. Watch cracking. Destroy stung fruit.	2.7
HARVEST 13–20 Oct	85–92	Harvest	Pick fully-ripe only (slip+net+aroma, brix ≥ 15). Sort, pack, sell now.	—

Greenhouse prep 21 Oct–7 Nov: remove, disinfect, flush, check equipment, re-buffer, make new stock.

Crop 3 — Idul Fitri (Dec 2026–Mar 2027)

Week	HST	Stage	Do this week	EC
Nursery 20–30 Nov	−14→0	Nursery	Soak 7–9 h → sprout → rockwool → bottom-feed. Harden. 2–3 true leaves.	0.5→1.5
1 1–7 Dec	0–6	Establish	**Transplant 1–5 Dec.** Settle roots. Water every 4 h→3 h. Train. Daily EC/pH.	2.1
2 8–14 Dec	7–13	Grow	Train vine. Remove laterals below node 8. Silicon drench weekly. Scout pests.	2.1
3 15–21 Dec	14–20	Grow	Keep training. Silicon weekly. Watch aphids/thrips/mildew.	2.1
4 22–28 Dec	21–27	Grow→Flower	First flowers (~HST 25). Feed EC 2.4 (Boroni in). Hand-pollinate 8–10 AM.	2.1→2.4
5 29 Dec–4 Jan	28–34	Flower/Set	Hand-pollinate daily 8–10 AM. Water every 2 h. Fruit-fly traps up.	2.4
6 5–11 Jan	35–41	Set/Select	**Keep ONE fruit/plant.** Top ~25 leaves. Start Ca foliar.	2.4
7 12–18 Jan	42–48	Sizing	Feed EC 2.55. Water every 2 h, 8 min. Support fruit. Ca foliar to HST 44.	2.55
8 19–25 Jan	49–55	Sizing	Netting forms. No leaf spray. Bag fruit/traps. Watch BER.	2.55
9 26 Jan–1 Feb	56–62	Sizing→Ripen	Fruit near full size. Prepare feed switch.	2.55→2.7
10 2–8 Feb	63–69	Ripen	Feed EC 2.7 (cut N, max K). Start brix testing. Steady water.	2.7
11 9–15 Feb	70–76	Ripen/Finish	Ripen feed continues. Rolling pick, ripe fruit only. Fruit fly CRITICAL.	2.7
12 16–23 Feb	77–84	Finish	Ripen feed. Brix ≥ 15. Watch cracking. Destroy stung fruit.	2.7
HARVEST 24 Feb–3 Mar	85–92	Harvest	**Harvest, sort & pack before 3 Mar** — market ≥1 wk before Idul Fitri. Ripe only.	—

Main rule — no delays

Crop 3 must be harvested, sorted & packed **before 3 March 2027** to reach market ≥1 week before Idul Fitri. Report any problem to management immediately.

Reference

Exact numbers for whoever makes the drums & checks the feed. Daily staff don't need this.

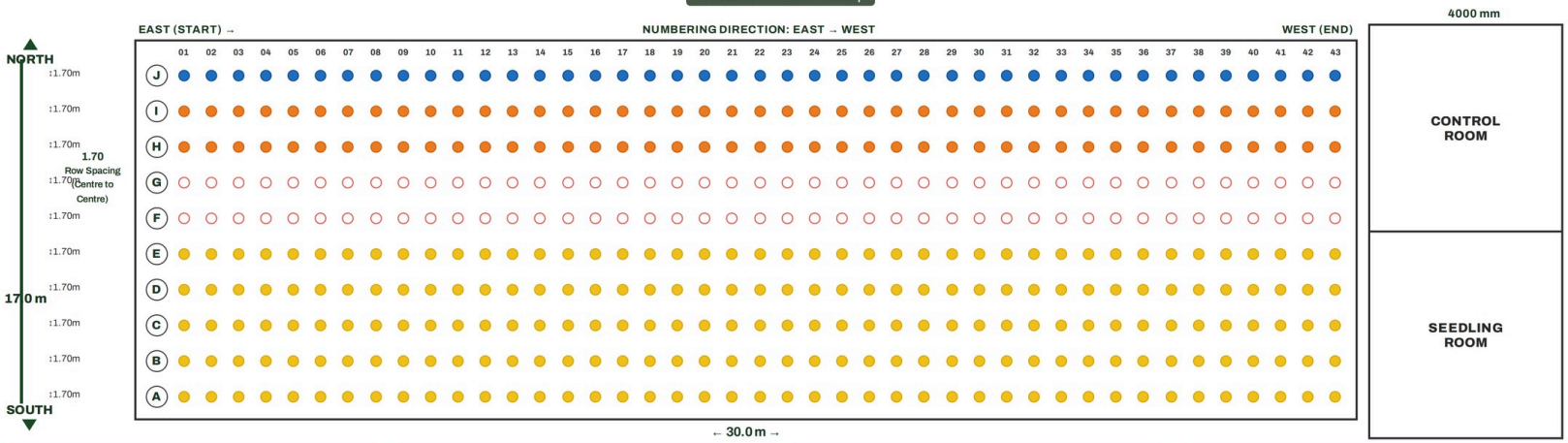
REFERENCE

Greenhouse map

SPARMANIK FARM · GREENHOUSE FLOOR PLAN

GREENHOUSE 1 – MAIN PRODUCTION HOUSE

ORIENTATION: NORTH ↑



VARIETY LEGEND

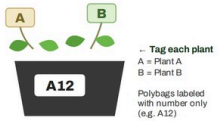
- **YELLOW KIRIN KEVIN**
  Rows A to E
  215 polybags • 430 plants • 430 seeds
- **WHITE KIRIN KEVIN**
  Rows F to G
  86 polybags • 172 plants • 172 seeds
- **SPARMANIK MANIS CANDY**
  (ORANGE CANDY SELECTION) · Rows H to I
  86 polybags • 172 plants • 172 seeds
- **YELLOW KIRIN AUSTRALIA F3**
  Row J
  43 polybags • 86 plants • 86 seeds

GREENHOUSE STATISTICS

Total Rows	10 (A–J)
Polybags per Row	43
Total Polybags	430
Plants per Polybag	2
Total Plants	860
Direction of Numbering	East → West
Row Order	A (South) to J (North)

🌱 Each plant is uniquely identified.
Format: **Row + Number + Plant (A/B)**
Example: A01A, A01B, A02A, A02B … J43A, J43B

IDENTIFICATION SYSTEM



← **Tag each plant**
A = Plant A
B = Plant B

Polybags labeled with number only (e.g. A12)

If one plant dies or has disease:

- Replace only that plant (e.g. A12B)
- The other plant (A12A) keeps its identity and record.

NOTES

- 2 seeds planted per polybag.
- Every plant has an individual ID for recording growth, health and production.
- This system makes selection, pollination and seed collection easier.
- Keep clean and sanitise to prevent infection.

SPACING

Polybag Spacing (Centre to Centre)	Row Spacing (Centre to Centre)	Plant Spacing (within Polybag)
625 mm	**1,700 mm**	**150 – 200 mm**

SEED REQUIREMENT

Seeds per Polybag	2
Seeds per Row	86
Total Seeds Required	860

Layout date: 27 May 2025 · Revision: all 10 rows A–J

Greenhouse 1 — Main Production House · 30.0 × 17.0 m growing area · Control room (Ruang Kontrol) & Seedling room (Ruang Seedling) at the east end · numbering South → North (rows), East → West (polybags).

REFERENCE

Greenhouse layout & plant ID

Greenhouse 1 — Main Production House is run as a **production & breeding facility**: every row, polybag and plant has a unique ID so any plant can be tracked seed-to-harvest.

Layout & rules

- **10 rows A–J**, numbered **South → North**; polybags numbered **01–43 East → West** per row.
- **43 polybags per row · 2 plants per polybag → 430 polybags, 860 plants** total.
- **One fruit per plant** — so 2 fruit per bag, each on its own plant.
- **Spacing:** rows 1.70 m (centre-to-centre) · polybags 625 mm · plants 150–200 mm within a bag.

Varieties by row

Rows	Variety	Polybags · seeds
A–E	Yellow Kirin Kevin	215 · 430
F–G	White Kirin Kevin	86 · 172
H–I	Sparmanik Manis Candy (Orange Candy Selection)	86 · 172
J	Yellow Kirin Australia F3	43 · 86

Plant ID system

Label each **polybag with its number only** (e.g. **A12**). Tag the two plants inside as **Plant A** and **Plant B** → full ID **A12A** and **A12B**. Format: *Row + Number + Plant* (A01A, A01B … J48A, J48B).

If a plant dies or gets sick

Replace **only that plant** (e.g. A12B). The other plant (A12A) **keeps its original ID and record**. Two plants in one bag perform differently — recording each one separately lets you pick the best genetics and cull the poor ones.

Records to keep (per plant)

- **Pollination:** log each hand-cross by plant, e.g. *H18A pollinated by H20B*.
- **Fruit quality:** weight, brix, shape, netting, skin colour, shelf life, disease resistance, yield.
- **Seed selection:** keep seed only from the best-performing plants.
- **Disease:** report by plant ID; check neighbours; record the treatment.

Staff rules

Always use the correct row + polybag + plant ID. **Never remove labels.** Report disease and record harvest against the right plant ID. Report any missing or damaged label immediately.

REFERENCE

Products

Product	Analysis	Role
Calcinit (Yara)	15.5% N · 19% Ca	Calcium, N — Tank A
Krista K Plus (KNO_3)	13.7% N · 38.4% K	Potassium, N — Tank A
Krista MKP	22.7% P · 28.2% K	Phosphorus — Tank B
Meroke MAG-S	9.6% Mg · 13% S	Magnesium — Tank B
Meroke SOP	~43% K · 18% S	Ripening K (no N) — Tank B
Vitaflex + Fe-EDDHA	Fe·Mn·Zn·Cu·B·Mo	Micros — Tank B
Karate Plus Boroni	Ca-nitrate + 0.3% B	Boron at flowering — Tank A (part-swap)
Kalsika	K-silicate	Silicon — root drench HST 8–36 only (4.0)

REFERENCE

Two tanks · stock · mixing

> ⚠ **Never mix calcium with sulphate/phosphate undiluted**
>
> Tank A = Calcinit · Boroni · K Plus. Tank B = MKP · MAG-S · Vitaflex+Fe. They meet only diluted in the 1000 L. **SOP is added separately** (below). Kalsika goes in neither.

Make the stock — large batch in a drum

Recipe key: S1 = Grow (Stage 3) · S2 = Flower (Stage 4) · S3 = Sizing (Stage 5) · S4 = Ripen (Stage 6). Staff pages use the stage & EC; only the mix room uses S1–S4.

5 L of stock feeds 1000 L. So a **50 L drum = ten 1000 L batches**; a **100 L drum = twenty**. Figures below are for a **50 L drum — double every number for a 100 L drum.** Don't make more than ~2 weeks' worth (a giant 1000 L stock drum would sit too long and the iron degrades). Make Tank A and Tank B in *separate* drums.

TANK A · kg / 50 L drum	Grow	Flower	Sizing	Ripen
Calcinit	8.7	9.45	9.45	8.95
Boroni	—	0.8	0.8*	—
K Plus	3.45	1.15	—	—

TANK B · kg / 50 L drum	Grow	Flower	Sizing	Ripen
MKP	2.2	2.3	2.1	1.85
MAG-S	5.7	6.05	6.25	6.25
Vitaflex	0.2	0.2	0.2	0.2
Fe-EDDHA	0.05	0.05	0.05	0.05

To fill a drum: part-fill with warm water, add one product at a time and stir until clear, then top up to the drum mark (50 L or 100 L). Take **5 L to each 1000 L feed tank**with a marked container. *4.0: no Boroni in S3. Keep drums warm; use within 1–2 weeks; keep out of sun. Never combine A & B undiluted.

SOP — separate, add straight into the 1000 L tank

SOP	Grow	Flower	Sizing	Ripen
Per 1000 L (one bag)	—	310 g	535 g	645 g
For a 50 L drum's 10 batches	—	3.1 kg	5.35 kg	6.45 kg

Why SOP is separate

Potassium sulphate barely dissolves (~110 g/L), so it's the one thing that *can't* go in a strong stock. Kept out of it, Stock A & B stay strong — only **5 L each** per 1000 L. Pre-weigh SOP into per-tank bags; pre-dissolve a bag in a bucket of warm water, then pour into the 1000 L (Flower / Sizing / Ripen only).

Full ingredient list for a drum batch

Tank A: **Calcinit, Boroni, K Plus.** Tank B: **MKP, MAG-S, Vitaflex, Fe-EDDHA.** Separate bag: **SOP.** Separate weekly drench (not in any drum): **Kalsika** (silicon).

Make the feed — per 1000 L

Per 1000 L	Grow	Flower	Sizing	Ripen
Stock A	5 L	5 L	5 L	5 L
Stock B	5 L	5 L	5 L	5 L
SOP bag	—	310 g	535 g	645 g
Set EC	2.0–2.2	2.3–2.5	2.5–2.6	2.6–2.8
Set pH	5.8–6.3	5.8–6.3	5.8–6.3	5.8–6.3

~900 L water → add 5 L A, stir → add 5 L B, stir → (Flower / Sizing / Ripen) tip in the SOP bag, stir → top to 1000 L → set EC then pH.

REFERENCE

PPM targets & ranges

The feed's designed output. Confirm by lab monthly (±10–15% is fine). Bold = target.

ppm	Grow	Flower	Sizing	Ripen
N	**180** 165–195	**175** 160–190	**158** 145–175	**138** 120–150
P	**50** 42–58	**52** 45–60	**48** 40–56	**42** 35–50
K	**195** 175–215	**240** 220–260	**285** 265–305	**325** 305–345
Ca	**165** 150–185	**195** 175–215	**195** 175–215	**170** 155–190
Mg	**55** 48–62	**58** 50–66	**60** 52–68	**60** 52–68

Micros (all stages): Fe 2.2 · Mn 0.30 · Zn 0.33 · B 0.2–0.44 (keep <0.6) · Cu 0.32 · Mo 0.05 ppm.

REFERENCE

Daily checks & log

Check	OK range	If out	✓
Feed EC	stage value	low→add A+B · high→dilute	☐
Feed pH	5.8–6.3	usually ~6.0 by itself; if >6.3 add acid	☐
Drain EC	≤ feed + 1.0	higher→flush more	☐
Drain %	15–20%	low→+1 min · high→–1 pulse	☐

Only change the timer at a stage change (or if run-off drifts for several days). Lab-test the feed monthly.

Feed lands at pH 6.0 on its own — usually no acid needed

Raw water is pH 7.7, but the fertilisers (especially MKP) pull the mixed feed down to about **pH 6.0** — right in the 5.8–6.3 target. So in normal running you **add no acid**; just confirm with the meter. Keep a little acid on hand only for the odd batch reading **above 6.3** — most likely in ripening (Sizing / Ripen, less MKP) or the dry season. Then add slowly to ~6.0 and note the mL.

Your water (tested)

pH ~7.6 · EC **0.18 mS/cm (184 µS)** · TDS ~86 ppm — soft and low in minerals, so alkalinity is low and the feed self-drops to ~6.0. Because the raw water is so low, nearly all the feed EC is your nutrients — the recipe delivers accurate ppm. **Read the feed in EC/mS mode** and aim for the stage EC. (If reading ppm/0.5 mode instead: ~1050 · 1200 · 1275 · 1350.) Calibrate both pens periodically.

Log

Date	Stage	Feed EC	pH	Drain EC	Drain %	By

REFERENCE

Watering timer (simple intervals)

Set the timer as a plain **every-2/3/4-hour** cycle for the stage, first pulse after sunrise, last well before dark. Small, frequent pulses — keep them running **through midday** (that's when the plant drinks most; don't skip the middle of the day).

Stage (HST)	Every	Min	Clock times	≈/day
Establish (0–7)	4 h	5	8am·12pm·4pm	0.5 L
Grow (8–25)	3 h	6	8am·11am·2pm·5pm	0.8 L
Flower/Set (25–40)	2 h	7	8am·10am·12pm·2pm·4pm	1.2 L
Sizing (40–60)	2 h	8	7am·9am·11am·1pm·3pm·5pm	1.6 L
Ripening (60–72)	3 h	6	8am·11am·2pm·5pm	0.8 L
Finish (72–harvest)	not used in 4.0 — Ripen feed to the last fruit			

Set once per stage

Enter the stage's interval & minutes, then leave it. Don't re-program for one rainy day — free-draining coir sheds the extra. Check run-off ~2–3×/week (aim 15–20%); only adjust the whole schedule if it drifts for several days. If the timer has no day/night setting, choose the start so pulses land in daylight — avoid watering into the night.

REFERENCE

Plant problem picture guide

Match what you see. **SEE → MEANS → DO.**



Brown sunken patch on fruit base

Calcium not reaching fruit (BER)

Airflow · steady water · ease early-K



Old leaves yellow between veins

Magnesium low

Check MAG-S · foliar Epsom 10 g/L



New leaves pale, veins green

Iron / pH too high

Lower pH ~6.0 · add Fe-EDDHA



Brown burnt leaf edges

Boron excess / salt high

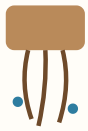
Check Boroni dose · check EC



White powder on leaves

Powdery mildew

Fans · morning water · fungicide · silicon



Coir always wet, wilting

Overwatering / root rot

Cut a pulse · add perlite · check drainage



Maggots / rot inside fruit

Fruit fly (lalat buah)

Traps · bag fruit · destroy stung fruit



Brown gummy ooze on stem base

Gummy Stem Blight (hot+humid)

Airflow · fungicide · remove infected · avoid stem wetness

REFERENCE

Greenhouse & climate

Your 30 × 17 m, 7 m tall house — net sides, double netted roof vent, fans 24/7, dry weed-mat floor, raised bags — is a complete passive-vent, low-condensation setup. In this mild, still, cloudy, ~90% climate you **can't reach low humidity**; the air just equals the outside.

✓ **So the goal isn't dry air**

It's: keep leaves dry & moving (fans), keep vents open & net clean, and win on **calcium** (generous Ca, moderate early-K, nitrate-N, steady water, preventive Ca foliar). Optional upgrade: powered gable exhaust fans (cut heat, not humidity below outside).

TEAR-OUT · PIN BY THE MIXING TANK

Quick card

Per 1000 L	Grow	Flower	Sizing	Ripen
Stock A	5 L	5 L	5 L	5 L
Stock B	5 L	5 L	5 L	5 L
SOP bag	—	310	535	645
EC	2.1	2.4	2.55	2.7
Water	3h·6′	2h·7′	2h·8′	3h·6′

Every day: 5 L A + 5 L B (+SOP bag: Flower/Sizing/Ripen) → 1000 L, set EC + pH 5.8–6.3 · check run-off.
Weekly:silicon HST 8–3**Always:** one fruit/plant · fans 24/7 · A & B never mixed undiluted.

Sparmanik Farm

SPARMANIK 4.0 · Melon SOP Book · 6 Oct 2026 · Kab. Simalungun, Sumatera Utara

Doses are validated starting points — your EC/pH meters, run-off and brix logs are the final authority. Confirm raw-water chemistry with a lab test each season.

KNOW YOUR PRODUCTS · 1 OF 2

What each product looks like

So nobody weighs the wrong bag. Photos from the store, Sep 2026.

Two colour rules

Yara sacks: read the fingerprint — Calcinit red, K Plus green, MKP navy. **Meroke packs:** all white with a red bird — read the droplet: MAG-S purple, Vitaflex pink, Fe dark maroon. **Boroni is the only orange sack.**



BAG A → DRUM A

YaraTera Calcinit

calcium nitrate · 15.5% N · 19% Ca

RED pattern. White prills. Pulls water from the air and sets hard — double-bag and seal.

25 kg sack · Rp 14,800/kg



BAG A → DRUM A

YaraTera Krista K Plus

potassium nitrate · 13.7% N · 38.4% K

GREEN pattern. Says KRISTA K PLUS. Grow and Flower only.

25 kg sack · Rp 46,000/kg

Not Krista MKP — that one is Drum B.



BAG A → DRUM A

Meroke Karate Plus Boroni

calcium nitrate · 15.5% N · 26% CaO · 0.3% B

ORANGE sack, sunburst — the only orange bag. 4.0: **Flower stock only**, in Boyd's kit.

25 kg sack · Rp 13,200/kg

Never boric acid.



BAG B → DRUM B

YaraTera Krista MKP

mono potassium phosphate · 22.7% P · 28.2% K

NAVY BLUE pattern. Says KRISTA MKP.

25 kg sack · Rp 63,000/kg

Not Krista K Plus — that one is Drum A.



BAG B → DRUM B

Meroke MAG-S

magnesium sulphate · 9.6% Mg · 13% S

PURPLE droplet. White crystals. Biggest weight in Drum B.

1 kg packs · Rp 16,000/kg

Not SOP — SOP goes in the tank.



BAG B → DRUM B

Meroke VITAFLEX

micros · Fe Mn Zn Cu B Mo

PINK droplet. Only 200 g per 50 L drum — kitchen scale.

500 g pack · Rp 200,000/kg



BAG B → DRUM B

Meroke MIKRO Fe 6%

EDDHA-Fe iron chelate

DARK MAROON droplet. Only 50 g per 50 L drum — kitchen scale.

500 g pack · Rp 273,904/kg

Must be EDDHA — never Fe-EDTA.

KNOW YOUR PRODUCTS · 2 OF 2

Tank, drench and spray products

None of these ever goes into Drum A or Drum B.



STRAIGHT INTO THE TANK

Meroke SOP

sulphate of potash · 43% K · 18% S

Purple panel, red bird. Dissolve in warm water first. Flower 310 g · Sizing 535 g · Ripen 645 g per tank.

1 kg packs · Rp 25,000/kg

Never in a drum. No SOP at Grow.



DRENCH · OWN PULSE

Kalsika

potassium silicate · 51.6% SiO_2

BLUE sachet, 100 g. One sachet in 300 L plain water. 4.0: HST 8, 15, 22, 29, 36 only.

100 g sachet · Rp 409,866/kg

Never after HST 36. 2 days from Trichoderma.



DRENCH · BY CUP

Tricoderma+ (Ruminesia)

Trichoderma harzianum + Bacillus, Azotobacter, Pseudomonas

Silver foil, yellow fruit label. 1 g per litre, 250 ml cup per bag. HST 0, 17, 38, 59.

1 kg pack · Rp 80,485/kg

Never in the drippers or feed tank.



FOLIAR · LEAF SPRAY

YaraVita Tri-Pholate

foliar micronutrient

Small white sachet, GREEN pattern. 0.125 g per litre (2 g per 16 L pack). HST 20, 28, 35, after 4 pm.

25 g pack · Rp 1,491,600/kg

Stop after HST 35 — marks the net.

The three that get confused

Krista MKP vs Krista K Plus — both Yara, both white, both say KRISTA. MKP → Drum B, K Plus → Drum A.

Calcinit vs Karate Plus Boroni — both calcium nitrate, both Drum A. Boroni carries boron: Flower stock only.

SOP vs MAG-S — both white 1 kg packs. SOP → straight into the tank. MAG-S → Drum B.

Banned from the farm

Boric acid (about 17.5% B, roughly 50× stronger than Boroni) · Fe-EDTA (stops feeding iron above pH 6.5) · Citric acid (made the water tower smell fishy; use pH Down 1 L in the mixed tank, only above pH 6.3).

WHAT WE PAY · 4.0

Cost per crop

Whole farm — 1,010 polybags · 2,020 plants · transplant to last fruit. Rates from price list v1.3 (Sep 2026).

17	19	26	18
GROW TANKS	FLOWER TANKS	SIZING TANKS	RIPEN TANKS (4.0)

Product	Rate / kg	v2.5 kg	v2.5 cost	4.0 kg	4.0 cost
Calcinit	Rp 14,800	66.27 kg	Rp 980,722	73.42 kg	**Rp 1,086,690**
MAG-S	Rp 16,000	43.69 kg	Rp 698,960	48.69 kg	**Rp 778,960**
SOP	Rp 25,000	26.25 kg	Rp 656,250	31.41 kg	**Rp 785,250**
Krista MKP	Rp 63,000	15.42 kg	Rp 971,460	16.90 kg	**Rp 1,064,700**
Krista K Plus	Rp 46,000	8.05 kg	Rp 370,300	8.05 kg	**Rp 370,300**
Karate Plus Boroni	Rp 13,200	3.60 kg	Rp 47,520	1.52 kg	**Rp 20,064**
Vitaflex	Rp 200,000	1.44 kg	Rp 288,000	1.60 kg	**Rp 320,000**
MIKRO Fe 6% EDDHA	Rp 273,904	360 g	Rp 98,605	400 g	**Rp 109,562**
Kalsika	Rp 409,866	800 g	Rp 327,893	500 g	**Rp 204,933**
Tricoderma+	Rp 80,485	1.01 kg	Rp 81,290	1.01 kg	**Rp 81,290**
Tri-Pholate	Rp 1,491,600	24 g	Rp 35,798	24 g	**Rp 35,798**
Total inputs			**Rp 4,556,798**		**Rp 4,857,547**

Rp 4,809	Rp 5,657,547	+Rp 300,748
PER POLYBAG · RP 2,405 PER PLANT	WITH LABOUR (RP 800,000)	4.0 VS V2.5

What changed from cost card v2.5

Ripen runs to the last fruit (4.0): 10 tanks to HST 72 plus about 8 more to HST 82, where v2.5 had free plain water. **Boroni** in Flower only: 1.52 kg, not 3.60 kg. **Kalsika** 5 rounds (HST 8–36), not 8. Tank counts and rates otherwise as v2.5.

Not included: seed, coir, electricity, water. Part-used tanks at a stage change can add up to 5% — 5-day batches (page 53) cut that.

NUTRITION · 4.0 · BATCH SIZE

Make 1, 5 or 10 days of stock

Same recipe, same strength — only the amount changes. Every 1,000 L tank still gets 5 L A + 5 L B.

1 day	5 days	10 days
5 L BUCKET A + B · 1 TANK	25 L DRUMS · NORMAL	50 L DRUMS · 10 TANKS

Batch	Use it when
1 day	Stage-change day · testing a fix · last days of the crop. Pour each **whole** bucket into the tank.
5 days	**Every week.** Fresh stock, little waste if the plan changes.
10 days	Only with 10+ days left in the stage and no change planned.

S1 GROW EC 2.1 · no SOP

Product	1 day · 5 L bucket	5 days · 25 L	10 days · 50 L
STOCK A · orange			
YaraTera Calcinit	870 g	**4.35 kg**	8.7 kg
Krista K Plus	345 g	**1.725 kg**	3.45 kg
Check weight A	**1.215 kg**	**6.075 kg**	**12.15 kg**
STOCK B · blue			
Krista MKP	220 g	**1.1 kg**	2.2 kg
Meroke MAG-S	570 g	**2.85 kg**	5.7 kg
Meroke Vitaflex	20 g	**100 g**	200 g
MIKRO Fe 6% EDDHA	5 g	**25 g**	50 g
Check weight B	**815 g**	**4.075 kg**	**8.15 kg**

NUTRITION · 4.0 · BATCH SIZE

Flower & Sizing batches

Under 1 kg → kitchen scale reading to 1 g. Same mixing method as page 15.

S2 FLOWER EC 2.4 · SOP 310 g / tank

Product	1 day · 5 L bucket	5 days · 25 L	10 days · 50 L
STOCK A · orange			
YaraTera Calcinit	945 g	**4.725 kg**	9.45 kg
Karate Plus Boroni (Boyd's kit)	80 g	**400 g**	800 g
Krista K Plus	115 g	**575 g**	1.15 kg
Check weight A	**1.14 kg**	**5.7 kg**	**11.4 kg**
STOCK B · blue			
Krista MKP	230 g	**1.15 kg**	2.3 kg
Meroke MAG-S	605 g	**3.025 kg**	6.05 kg
Meroke Vitaflex	20 g	**100 g**	200 g
MIKRO Fe 6% EDDHA	5 g	**25 g**	50 g
Check weight B	**860 g**	**4.3 kg**	**8.6 kg**

S3 SIZING EC 2.55 · SOP 535 g / tank · no Boroni

Product	1 day · 5 L bucket	5 days · 25 L	10 days · 50 L
STOCK A · orange			
YaraTera Calcinit	945 g	**4.725 kg**	9.45 kg
Check weight A	**945 g**	**4.725 kg**	**9.45 kg**
STOCK B · blue			
Krista MKP	210 g	**1.05 kg**	2.1 kg
Meroke MAG-S	625 g	**3.125 kg**	6.25 kg
Meroke Vitaflex	20 g	**100 g**	200 g
MIKRO Fe 6% EDDHA	5 g	**25 g**	50 g
Check weight B	**860 g**	**4.3 kg**	**8.6 kg**

NUTRITION · 4.0 · BATCH SIZE

Ripen batch & kits

Ripen has its own drums (v3.8 page 30) — not Sizing drums with extra SOP.

S4 RIPEN EC 2.7 · SOP 645 g / tank

Product	1 day · 5 L bucket	5 days · 25 L	10 days · 50 L
STOCK A · orange			
YaraTera Calcinit	895 g	**4.475 kg**	8.95 kg
Check weight A	**895 g**	**4.475 kg**	**8.95 kg**
STOCK B · blue			
Krista MKP	185 g	**925 g**	1.85 kg
Meroke MAG-S	625 g	**3.125 kg**	6.25 kg
Meroke Vitaflex	20 g	**100 g**	200 g
MIKRO Fe 6% EDDHA	5 g	**25 g**	50 g
Check weight B	**835 g**	**4.175 kg**	**8.35 kg**

Best practice: pre-weighed kits

Boyd weighs one bag per drum per batch, labelled e.g. “Ripen · A · 5 days”. Staff tip the bag in, top up to the 25 L line and stir until clear. Boroni is only ever in a kit.

Small amounts matter

In a 1-day bucket Fe-EDDHA is 5 g and Vitaflex 20 g — 1 g wrong is 20% wrong. Use the kitchen scale, or a kit from Boyd.

CLIMATE · 4.0

Humidity ≥ temperature × 2

v3.8 planned for a humid house. September 2026 showed the other risk: hot, dry afternoons.

Below this line the air is too dry: leaves close, stop making sugar and pull water so hard that fruit splits. The rule keeps VPD at about 0.8–1.6 kPa.

Temperature	Minimum humidity — below: mist leaves	Too wet — above: stop, open vents
28 °C	**56%**	79%
30 °C	**60%**	81%
32 °C	**64%**	83%
34 °C	**68%**	85%
36 °C	**72%**	87%

Every day with the Krisbow meter

1. Read and photograph it at **08:00 · 10:00 · 12:00 · 14:00**. The 12:00–14:00 readings matter most.
2. Too dry → mist the **leaves only** with the spray pack, 5–10 minutes. Check again after 30 minutes. Never the fruit, never wet the bags.
3. Over 80% → stop misting, all vents open, fans on.
4. Last misting **15:30**, so leaves are dry before night.
5. At night open vents and doors where possible: cool nights keep sugar in the fruit.

2026 readings

21 Sep: 34.8 °C at 59% — VPD about 2.3 kPa, far too dry; fruit had split the day before.
25 Sep 10:30: 31.7 °C at 71.8% — VPD about 1.3 kPa, ideal.

HARVEST · 4.0

Rolling harvest & grading

Pick only ripe fruit every morning; leave the rest on the Ripen feed to sweeten.

Brix 15+	1.8–2.2 kg	06:00–09:30
PASS	TARGET WEIGHT	PICK WHILE COOL

Every morning — walk every row

Fruit	Action
Split or rotten	Cut off → sack → **out of the greenhouse**. Count.
Ripe: full net · ground spot yellow · aroma · slips with a gentle twist	**Pick** with scissors, short T-stalk. Shade, one layer. Count + weigh.
Everything else	**Don't cut.** Leave it until it is ripe.

Brix test without wasting fruit

From HST 68 test fruit that **split that day**, flesh still clean. Juice from mid-flesh, not the seed cavity. Never cut a good fruit to test. Old splits and brown flesh read wrong.

Grading A–D

Grade	Weight	Brix	Skin	Goes to
A	1.8 – 2.2 kg	15+	Clean, full net	Premium
B	1.2 – 1.8 or over 2.2 kg	14+	Clean or minor marks	Second grade
C	Under 1.2 kg	13+	Marks, sound flesh	Local sale
D	—	—	Split, rotten, brown flesh, fly-stung	Destroy outside

Don't wash. Log each fruit against its plant ID. Tank and timer stay on Ripen until the last fruit is picked.

WEEKLY CHECKS · EVERY MONDAY

Is the crop on track?

Measure the same 5 marked plants — one per variety block plus one at a row end. Two checks in a row behind → call Boyd.

HST	Milestone	Height (guide)	Leaves (guide)	Ours
0	Transplant: 2–3 true leaves, sturdy, even	8–12 cm	2–3	
7	Start removing side shoots (nodes 1–8)	15–20 cm	4–6	
14	Vine climbing its string	35–50 cm	7–9	
20	Node 8 — stop removing shoots	70–100 cm	10–12	
25	First female flowers, nodes 9–12	110–140 cm	14–16	
38–40	One fruit per plant; top at leaf 25	180–200 cm	25	
45	Netting starts	—	25	
55	Fruit about 1.2 kg	—	23+	
60	Near full size, netting complete	—	22+	
68	Start Brix tests	—	20+	
75–85	Harvest 1.8–2.2 kg, Brix 15+	—	20+	

Heights and leaf counts are guides until we have our own averages — write ours in and they become the 4.1 numbers. 2026: 786 g average at HST 51 was the early sign the crop was behind.

Topping at leaf 25 (HST 38–46)

Plant has	Action
A fruit + 25 or more green leaves	**Top just above leaf 25**
A fruit + fewer than 25	Not yet
No fruit	**Don’t top** — it can never flower again

Count, don’t guess. A leaf with brown edges and a green middle still counts.

Fruit weight log — from HST 42, 10 fruit per variety

Date	HST	Yellow Kevin	White Kevin	Manis Candy	Australia F3	Name

RUN-OFF · CALL BOYD

Run-off in millilitres & when to call

Measure what drains from one marked bag: bucket under it in the morning, jug in the evening.

Stage	HST	Water per bag	Target	Run-off ml
Establish	0–7	0.5 L	15–20%	**75–100**
Grow	8–24	0.8 L	15–20%	**120–160**
Flower	25–41	1.2 L	15–20%	**175–235**
Sizing	42–59	1.6 L	15–20%	**240–320**
Ripen	60+	0.8 L	10–15%	**80–120**

Low for several days → add 1 minute to every pulse. High → remove 1 pulse. Never change it for one day or rain.

Call Boyd the same day when

- Feed EC is off the stage number, or pH outside 5.8–6.3, after mixing — **pump stays off**.
- Run-off EC more than 1,000 above the feed (e.g. over 3,400 at Flower), or run-off pH over 6.8.
- Run-off out of range three days running.
- More than 50 wet cracks, or splitting two days running.
- A dry bag not recovered overnight, or three or more bags wilting.
- Deformed fruit, dead flowers, or burnt leaf edges.
- Gummy ooze on a stem, or the first mildew spots.
- Humidity still under temperature × 2 after misting.
- Two weekly checks in a row behind.
- Anything you are not sure about. Stop and ask — don’t guess.

EMERGENCY ONLY · BOYD DECIDES

Feed-mistake flush

Used once in 2026 (9 Sep, HST 41) after boric acid went in. Never a routine job.

Only Boyd starts a flush

A flush is plain water at the roots. Done at the wrong time it splits fruit. Use it only to wash out a feed mistake — wrong sack, wrong dose.

1. Mark 5 bags (different rows, one at a row end). Measure their run-off EC **before** the flush.
2. Tank 1: 1,000 L plain water at pH 6. No fertiliser. Run the drippers **30 minutes** (about 1 L per bag).
3. While it runs, hand-pour **0.5 L round the edge** of each bag — the dripper only wets the middle.
4. Tank 2: refill, run 30 minutes, keep pouring edges. Tank 3: refill, run 30 minutes. Three tanks = about 3 L per bag.
5. Push a finger into the middle and the edge of 10 bags. Still dry inside → 2 L by hand on that bag only.
6. Measure the **last** run-off from the same 5 bags and send the numbers to Boyd.
7. **The same afternoon** remake the tank (EC 1300–1400 first, then the stage feed) and water normally. Bags must never sit in plain water overnight.

After the flush

Keep pollinating every new flower — damaged pollen means fewer set fruit. Expect the crop to run about 5 days behind; Boyd moves every later date by the same amount.

PROBLEMS · 2026 LESSONS

What went wrong in 2026 — and the fix

Add these to the picture guide on page 47.

What you see	Cause in 2026	Do	Call Boyd when
Many fresh wet splits at once	Plain-water drench, glass watering, dry air	Remove split fruit; check nobody added water; mist leaves	> 50 cracks, or 2 days running
Old dry corky cracks, brown patches	Old boron damage	Leave dry ones; cut wet or smelly ones	Brown flesh in sound fruit
One bag wilting, coir dry inside	Blocked dripper / water-repellent coir	Soak that bag 30–60 min in plain water, clear the dripper	Not recovered next morning, or 3+ bags
New leaves pale, green veins	Fe-EDTA at high run-off pH	Run-off pH weekly; Fe-EDDHA only	Run-off pH over 6.8
Deformed fruit, dead flowers	Boric acid instead of Boroni	Keep pollinating new flowers; check the sack in Stock A	Always, same day
Water tower smells fishy	Citric acid fed bacteria	Drain and rinse the tower; pH Down only, in the mixed tank, only above 6.3	Always, same day
Whole house wilting at midday	34.8 °C, 59% humidity	Mist leaves, shade, vents (page 56)	Still wilting at 16:00
Ripe-looking fruit, Brix under 12	Not ripe, or damaged	Leave that row longer	Main harvest within 3 days

Never (added in 4.0)

- Boric acid, Fe-EDTA or citric acid anywhere on the farm.
- Plain water, glass watering or silicon after fruit set (HST 38).
- MKP and Calcinit in the same spray pack. Leaf sprays HST 45–59.
- Cutting a good fruit to test Brix.

RECORD · SEASON 2026

2026 crop & the next transplant

Transplant 30 Jul 2026 (HST 0) · main harvest 3 Oct (HST 65).

19.0	17.4	15.5	14.6
WHITE KIRIN	AUSTRALIA F3	MANIS CANDY	YELLOW KIRIN

Cracking stopped once EC was held steady with no plain water. Fruit stayed small (0.7–1.5 kg) from the boric acid, the boron flush, Fe-EDTA iron lock-up, heat and a Sizing switch 8 days late.

Date	HST	What happened
3 Oct	65	Main harvest; rolling pick before and after
29 Sep	61	Ripen EC 2700, MKP spray, glass watering stopped
25 Sep	57	Silicon cancelled; tank 2505 / pH 5.8
20–21 Sep	52–53	Splitting after the 19 Sep silicon; a Kirin row lost; 34.8 °C / 59%
19 Sep	51	514 fruit weighed, average 786 g
18 Sep	50	Sizing EC 2550 started (8 days late)
9 Sep	41	Boron flush, back to Flower 2400 without boron
before 8 Sep	—	Boric acid used instead of Boroni
30 Jul	0	Transplant; Grow stock made with Fe-EDTA 13%

Before the next transplant

- ☐ Staff walked through the 4.0 changes (front page)
- ☐ Boric acid and Fe-EDTA removed from the farm
- ☐ Raw water and feed lab-tested
- ☐ Coir washed, rested and re-buffered (page 6)
- ☐ Pre-weighed 5-day kits for Grow and Flower ready
- ☐ Tri-Pholate, Tricoderma+ and Kalsika in stock (page 52)
- ☐ Krisbow, refractometer, EC and pH pens calibrated
- ☐ Fruit-fly and sticky traps in stock